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Rigidity at the Extremal Crouzeix Constant

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Abstract

For a complex matrix A, let $\psi(A)$ be the norm of the polynomial functional calculus on its numerical range. We prove that $\psi(A) = 2$ only when the numerical range is a nondegenerate closed disk. The proof first extracts the equality conditions in the sharp double-layer estimate. After boundary eigenvalues are removed, an extremizer obtained from a finite Blaschke product satisfies an exact boundary-kernel identity. On a curved exposed arc this identity makes the extremizer rational. A direct resolvent argument then places every zero of its reduced numerator inside the numerical range and every pole outside; consequently its full unit lemniscate coincides with the boundary of the numerical range. The defining polynomial of the projective dual of that algebraic boundary divides the Hermitian Kippenhahn determinant. Hyperbolicity and strict convexity force the dual curve to have degree two, and its points at infinity force the resulting conic to be a circle.

Keywords: numerical range; Crouzeix constant; spectral set; rational lemniscate; hyperbolic polynomial

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1. Introduction

Throughout, $\mathbb{C}$ and $\mathbb{R}$ denote the complex and real fields, $M_N(\mathbb{C})$ the algebra of $N \times N$ complex matrices, and I the identity matrix of the size dictated by context. For a matrix B, the symbols B^*, $\operatorname{spec}(B)$, and $\operatorname{spr}(B)$ denote its conjugate transpose, spectrum, and spectral radius. The notation $\|\cdot\|$ denotes the Hilbert-space norm on vectors and the induced operator norm on bounded linear maps; on $\mathbb{C}^N$ these are the Euclidean and induced Euclidean norms. We put

$$\operatorname{Re} B = \frac{B + B^*}{2}, \qquad \operatorname{Im} B = \frac{B - B^*}{2i}.$$

For scalars and functions, Re and Im have their usual pointwise meanings. For Hermitian matrices B, C, the relation $B \succeq C$ means that $B - C$ is positive semidefinite. We write $\operatorname{int} E$ for the interior of a planar set E and $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ for the open unit disk. Scalar matrix coefficients are written as $x^* B y = \langle By, x \rangle$, where inner products are linear in the first argument. For a bounded scalar-valued function g on a set E, write $\|g\|_E = \sup_{z \in E} |g(z)|$.

For $A \in M_N(\mathbb{C})$, set

$$W(A) = \{u^* A u : u \in \mathbb{C}^N,\ \|u\| = 1\}.$$

For a polynomial p, put

$$\psi(A) = \sup\Big\{ \|p(A)\| : p \in \mathbb{C}[z], \|p\|_{W(A)} \le 1 \Big\}.$$

For a compact convex body K, let $C(K)$ denote the complex-valued continuous functions on K, and write

$$\mathcal{A}(K) = \{ f \in C(K) : f|_{\operatorname{int} K} \text{ is holomorphic} \}.$$

The space $H^\infty(\operatorname{int} K)$ consists of the bounded holomorphic functions on $\operatorname{int} K$ and carries the norm $\|\cdot\|_{\operatorname{int} K}$. Whenever $\operatorname{spec}(A) \subset \operatorname{int} K$, the notation $f(A)$ denotes the standard finite-dimensional holomorphic functional calculus, equivalently evaluation of any Hermite interpolating polynomial with the required spectral jets. The constant $\psi(A)$ was introduced by Crouzeix [1], who proved $\psi(A) \le 2$ for 2×2 matrices and conjectured the same bound in every dimension. He subsequently obtained the first dimension-free bound [2]. The disk case is the classical Okubo–Ando theorem [3], while the positive operator-valued double-layer representation behind the general convex-domain estimates goes back to Delyon and Delyon [4]. Crouzeix and Palencia reduced the universal constant to $1 + \sqrt{2}$ [5].

Jin [6] and, independently, Lorist and Schwenninger [7] proved the scalar conjecture. The latter proof applies an abstract 2-dilation perturbation lemma simultaneously to all iterates in the double-layer representation. Subsequently, Åhag, Czyż, Perälä, and Virtanen established sharp square-function inequalities, the complete conjecture in dimension three, and equality, stability, and representing-measure results [8]. Their equality theory also produces a two-dimensional reducing nilpotent block and strong dimension-three restrictions.

This paper develops the equality data into an all-dimensional geometric rigidity theorem. Its central steps are rational collapse of the extremizer, identification of the full lemniscate with the numerical-range boundary, and a hyperbolic projective-dual contact count.

Theorem 1. *For every $N \ge 1$ and $A \in M_N(\mathbb{C})$,*

$$\psi(A) = 2 \quad \Longrightarrow \quad W(A) = \{z \in \mathbb{C} : |z - \gamma| \le \rho\}$$

for some $\gamma \in \mathbb{C}$ and some $\rho > 0$.

The proof has four steps. An equality analysis of the sharp bound produces two extremal vectors and a pointwise boundary-kernel identity. A curved exposed arc then turns that identity into a rational formula for the extremizer. A direct resolvent argument identifies its full unit lemniscate with $\partial W(A)$. Finally, the Kippenhahn determinant contains the dual curve as a hyperbolic factor. A generic contact count makes the dual a conic, and the circular points at infinity then force $\partial W(A)$ to be a circle. A separate ingredient, strict domain monotonicity of the matrix H^∞ functional-calculus norm, rules out a proper disk produced by the boundary-spectrum induction. Extensive numerical optimization by Greenbaum and Overton [10] repeatedly found stationary configurations with ratio 2 whose numerical ranges were disks; the theorem explains that pattern at exact equality.

2. Numerical-range preliminaries

Here and below, $\operatorname{conv} E$ denotes the convex hull of E. For a Hermitian matrix B, $\lambda_{\max}(B)$ denotes its largest eigenvalue. At a differentiability point σ of a convex boundary, $\mathfrak{n}(\sigma)$ denotes its outward unit normal; ν, with or without a subscript, denotes an arbitrary supporting unit normal.

Lemma 1 (Numerical-range geometry)**.** *The following facts hold.*

1. *$W(A)$ is nonempty, compact, and convex.*
2. *If $W(A)$ has empty interior, then A is normal and $\psi(A) = 1$.*
3. *For $\alpha \in \mathbb{C} \setminus \{0\}$, $\beta \in \mathbb{C}$, and unitary $\mathcal{U}$,*

$$W(\alpha A + \beta I) = \alpha W(A) + \beta, \qquad \psi(\alpha A + \beta I) = \psi(A), \qquad \psi(\mathcal{U}^* A \mathcal{U}) = \psi(A).$$

Also $W(A^*) = \overline{W(A)}$ and $\psi(A^*) = \psi(A)$.

4. *For a finite nonempty direct sum,*

$$W\left(\bigoplus_j A_j\right) = \operatorname{conv}\Big(\bigcup_j W(A_j)\Big), \qquad \psi\left(\bigoplus_j A_j\right) \le \max_j \psi(A_j).$$

5. *The support function of $W(A)$ is*

$$h_A(\theta) = \lambda_{\max}\Big(\operatorname{Re}(e^{-i\theta}A)\Big).$$

Proof. The unit sphere is compact and $u \mapsto u^*Au$ is continuous, proving nonemptiness and compactness. To prove convexity, take two points of $W(A)$ and compress A to the span of corresponding unit vectors. If the points are distinct, those vectors are linearly independent; if they coincide, there is no segment to prove. It is therefore enough to know that the numerical range of a 2×2 matrix is convex. After unitary triangularization such a matrix has the form $\left(\begin{smallmatrix} \lambda_1 & \tau \\ 0 & \lambda_2 \end{smallmatrix}\right)$. For $u = (\cos t, e^{i\varphi}\sin t)$, where $0 \le t \le \pi/2$ and $\varphi \in \mathbb{R}$,

$$u^*Au = \lambda_1 \cos^2 t + \lambda_2 \sin^2 t + \tau e^{i\varphi} \sin t \cos t.$$

These points fill the possibly degenerate ellipse with foci λ_1, λ_2, hence a convex set. The two-dimensional compression is contained in $W(A)$, so the segment joining the original two points is contained in $W(A)$.

If $W(A)$ has empty interior, its convexity places it in an affine line. If $W(A) = \{\gamma\}$, then $x^*(A - \gamma I)x = 0$ for all x. The complex polarization identity makes every matrix coefficient of $A - \gamma I$ zero. If $W(A)$ lies in a line, a translation and a nonzero complex rotation make all quadratic forms of A real. The skew-Hermitian part then has zero quadratic form and vanishes by polarization, so the transformed matrix is Hermitian. A normal matrix satisfies

$$\|p(A)\| = \|p\|_{\operatorname{spec}(A)} \le \|p\|_{W(A)}.$$

The constant polynomial $p \equiv 1$ gives the reverse inequality.

The affine and unitary formulas follow by changing variables in polynomials. For adjoints use $\widetilde{p}(z) = \overline{p(\overline{z})}$, for which $\widetilde{p}(A^*) = p(A)^*$. For a direct sum, a unit vector decomposes into block components and its numerical value is a convex combination of block numerical values. Also $p(\bigoplus_j A_j) = \bigoplus_j p(A_j)$; the supremum norm on the convex hull dominates that on every block. Finally,

$$\max_{z \in W(A)} \operatorname{Re}(e^{-i\theta}z) = \max_{\|u\|=1} u^* \operatorname{Re}(e^{-i\theta}A)u,$$

which is the largest eigenvalue of the displayed Hermitian matrix. □

3. An equality-preserving sharp-bound argument

We use the equality-preserving form of the Lorist–Schwenninger iterate argument [7], which records the pointwise identities needed for rigidity.

Lemma 2 (An abstract dilation estimate). *Let T act on a nonzero finite-dimensional Hilbert space. Suppose that $\mathcal{M}$ is a contraction on another Hilbert space, $\mathcal{V}$ is an isometry, and*

$$E_n = 2\mathcal{V}^*\mathcal{M}^{*n}\mathcal{V} - T^{*n} \qquad (n \ge 1)$$

are uniformly bounded and commute with T. Then $\|T\| \le 2$.

If additionally $\|T\| = 2$, $\operatorname{spr}(T) < 1$, $T^*Tx = 4x$, $\|x\| = 1$, and $y = Tx/2$, then

$$T^*x = 0, \quad T^*y = 2x, \quad \mathcal{M}\mathcal{V}x = \mathcal{V}y, \quad \mathcal{M}^*\mathcal{V}y = \mathcal{V}x.$$

Proof. Put $\kappa = \|T\|$. If $\kappa \le 1$, the conclusion is immediate; assume $\kappa > 1$. Choose a unit vector x with $T^*Tx = \kappa^2 x$, and set

$$m_n = \operatorname{Re}\big(x^* E_n T^n x\big), \qquad \omega = \mathcal{M}^*\mathcal{V}Tx - \kappa\mathcal{V}x.$$

The defining identity gives $\|T^{*n}\| \le 2 + \sup_k \|E_k\|$, so the powers of T, and hence (m_n), are bounded. Put $Q_n = \mathcal{V}^*\mathcal{M}^{*n}\mathcal{V}$ and $q_n = T^{*n}x$. Commutativity of E_n with T gives

$$2(Q_{n+1}T - TQ_{n+1}) = T^{*(n+1)}T - TT^{*(n+1)}.$$

Moreover, $m_n = 2\operatorname{Re}(q_n^* Q_n x) - \|q_n\|^2$ and $\mathcal{V}^*\mathcal{M}^{*n}\omega = Q_{n+1}Tx - \kappa Q_n x$. Multiplying the displayed commutator identity on the left by q_n^*, using $T^*Tx = \kappa^2 x$, and taking real parts now gives

$$\begin{aligned}
\kappa m_n - m_{n+1} &= \kappa(\kappa-1)\|T^{*n}x\|^2 - 2\operatorname{Re}\big((\mathcal{V}^*\mathcal{M}^{*n}\omega)^* T^{*n}x\big) \\
&= \kappa(\kappa-1)\left\| T^{*n}x - \frac{\mathcal{V}^*\mathcal{M}^{*n}\omega}{\kappa(\kappa-1)} \right\|^2 - \frac{\|\mathcal{V}^*\mathcal{M}^{*n}\omega\|^2}{\kappa(\kappa-1)} \\
&\ge -\frac{\|\omega\|^2}{\kappa(\kappa-1)}.
\end{aligned}$$

Telescoping, with the boundedness of m_n, yields

$$\kappa m_1 = \sum_{n\ge 1} \kappa^{-n+1}(\kappa m_n - m_{n+1}) \ge -\frac{\|\omega\|^2}{(\kappa-1)^2}. \tag{3.1}$$

On the other hand, using that $\mathcal{V}$ is isometric and $\mathcal{M}$ contractive,

$$\begin{aligned}
\|\omega\|^2 &\le 2\kappa^2 - 2\kappa\operatorname{Re}\big(x^*\mathcal{V}^*\mathcal{M}^*\mathcal{V}Tx\big) \\
&= 2\kappa^2 - \kappa m_1 - \kappa^3.
\end{aligned} \tag{3.2}$$

Combining (3.1) and (3.2) gives

$$\|\omega\|^2\left(1 - \frac{1}{(\kappa-1)^2}\right) \le \kappa^2(2-\kappa),$$

which is impossible for $\kappa > 2$.

Now let $\kappa = 2$. Equations (3.1)–(3.2) become $2m_1 \ge -\|\omega\|^2$ and $\|\omega\|^2 \le -2m_1$, so equality holds in both. For each n, the slack in the estimate preceding (3.1) is

$$2\left\| T^{*n}x - \tfrac{1}{2}\mathcal{V}^*\mathcal{M}^{*n}\omega \right\|^2 + \tfrac{1}{2}\big(\|\omega\|^2 - \|\mathcal{V}^*\mathcal{M}^{*n}\omega\|^2\big).$$

Their positively weighted sum is zero. Thus, for every n,

$$\mathcal{V}^*\mathcal{M}^{*n}\omega = 2T^{*n}x, \qquad \|\mathcal{V}^*\mathcal{M}^{*n}\omega\| = \|\omega\|.$$

Since $\operatorname{spr}(T) < 1$, $T^{*n}x \to 0$, and these two equalities force $\omega = 0$. Their first instance then gives $T^*x = 0$. With $y = Tx/2$, $\mathcal{M}^*\mathcal{V}y = \mathcal{V}x$. Both $\mathcal{V}x$ and $\mathcal{V}y$ are unit vectors, so equality in Cauchy–Schwarz for $\mathcal{M}$ gives $\mathcal{M}\mathcal{V}x = \mathcal{V}y$. Finally $T^*y = T^*Tx/2 = 2x$. □

Proposition 1 (The sharp numerical-range estimate)**.** *For every $N \ge 1$, every $A \in M_N(\mathbb{C})$, and every polynomial p,*

$$\|p(A)\| \le 2\|p\|_{W(A)}. \tag{3.3}$$

More generally, if K is a compact convex body, $W(A) \subset K$, $\operatorname{spec}(A) \subset \operatorname{int} K$, and $f \in \mathcal{A}(K)$, then

$$\|f(A)\| \leq 2\|f\|_K. \tag{3.4}$$

For $f \in H^\infty(\operatorname{int} K)$, the corresponding estimate is

$$\|f(A)\| \leq 2\|f\|_{\operatorname{int} K}.$$

Proof. Normalize $\|f\|_K \leq 1$. Cauchy's boundary formula below also holds when f is only in $\mathcal{A}(K)$: for $z_0 \in \operatorname{int} K$, the functions $f(z_0 + r(z - z_0))$, $r < 1$, are holomorphic on a neighborhood of K and converge uniformly to f; apply the usual formula to them and pass to the limit. The positive operator-valued double-layer kernel used below originates in Delyon and Delyon [4] and appears in the Cauchy-transform form of Crouzeix and Palencia [5]; see also the recent double-layer and configuration-constant treatments in [11,12]. Parametrize the rectifiable convex Jordan curve $\Gamma = \partial K$ counterclockwise by arclength. Its outward unit normal $\mathfrak{n}(\sigma)$ exists almost everywhere and has a measurable representative. All boundary identities below are understood arclength-almost everywhere. Put $R_\sigma = (\sigma I - A)^{-1}$ and

$$\mathcal{P}(\sigma) = \frac{1}{\pi} \operatorname{Re}\big(\mathfrak{n}(\sigma) R_\sigma\big).$$

At a point with a normal, the supporting-half-plane property gives

$$\Delta_\sigma := \operatorname{Re}\big(\overline{\mathfrak{n}(\sigma)}(\sigma I - A)\big) \succeq 0.$$

Multiplication by the two resolvents shows

$$\operatorname{Re}\big(\mathfrak{n}(\sigma) R_\sigma\big) = R_\sigma^* \Delta_\sigma R_\sigma \succeq 0. \tag{3.5}$$

Since $d\sigma = i\mathfrak{n}(\sigma)\, ds$ and $d\overline{\sigma} = -i\overline{\mathfrak{n}(\sigma)}\, ds$,

$$\mathcal{P}(\sigma)\, ds = \frac{1}{2\pi i} R_\sigma\, d\sigma - \frac{1}{2\pi i} R_\sigma^*\, d\overline{\sigma}.$$

Cauchy's formula gives $\int_\Gamma R_\sigma\, d\sigma = 2\pi i I$; taking adjoints gives $\int_\Gamma R_\sigma^*\, d\overline{\sigma} = -2\pi i I$. Therefore

$$\int_\Gamma \mathcal{P}(\sigma)\, ds = 2I. \tag{3.6}$$

The matrix field $\mathcal{P}$ is measurable and positive semidefinite almost everywhere. Since the positive-square-root map is continuous on the cone of positive semidefinite matrices, $\sigma \mapsto \mathcal{P}(\sigma)^{1/2}$ is measurable (with arbitrary values on the exceptional null set). Consequently

$$(\mathcal{V}x)(\sigma) = 2^{-1/2} \mathcal{P}(\sigma)^{1/2} x$$

defines an isometry into $L^2(\Gamma; \mathbb{C}^N)$. Let M_f be multiplication by f, and set $T = f(A)$. For every $n \geq 1$, Cauchy's formula and the preceding decomposition give

$$2\mathcal{V}^* M_f^{*n} \mathcal{V} = \frac{1}{2\pi i} \int_\Gamma \overline{f(\sigma)^n} R_\sigma\, d\sigma + T^{*n},$$

because

$$\int_\Gamma \overline{f(\sigma)^n} R_\sigma^*\, d\overline{\sigma} = \left(\int_\Gamma f(\sigma)^n R_\sigma\, d\sigma \right)^* = -2\pi i T^{*n}.$$

Thus

$$E_n := 2\mathcal{V}^* M_f^{*n} \mathcal{V} - T^{*n} = \frac{1}{2\pi i} \int_\Gamma \overline{f(\sigma)^n} R_\sigma\, d\sigma. \tag{3.7}$$

The right side commutes with T, because every resolvent does. Also

$$\sup_n \|E_n\| \le \frac{\mathrm{length}(\Gamma)}{2\pi} \max_{\sigma\in\Gamma} \|R_\sigma\| < \infty.$$

Lemma 2 proves (3.4).

For $f \in H^\infty(\operatorname{int} K)$, normalize $\|f\|_{\operatorname{int} K} \le 1$, choose a Riemann map $\phi : \operatorname{int} K \to \mathbb{D}$, and put $g = f \circ \phi^{-1}$. For $0 < r < 1$, define

$$f_r(z) = g(r\phi(z)).$$

The Carathéodory extension of ϕ shows that $f_r \in \mathcal{A}(K)$, and $\|f_r\|_K \le 1$. Moreover, $g(r\,\cdot) \to g$ locally uniformly with all derivatives. The finitely many spectral jets therefore converge, so $f_r(A) \to f(A)$; applying (3.4) and passing to the limit proves the H^∞ estimate.

For an arbitrary matrix A, apply (3.4) to the outer parallel convex body

$$K_\varepsilon = W(A) + \varepsilon\overline{\mathbb{D}},$$

which contains $W(A)$ in its interior. For every polynomial p, uniform continuity on a fixed compact neighborhood gives $\|p\|_{K_\varepsilon} \to \|p\|_{W(A)}$. Letting $\varepsilon \downarrow 0$ proves (3.3), including the boundary-spectrum and nonsmooth cases. □

Proposition 2 (Equality data)**.** *Let $K = W(A)$ have nonempty interior, let $\operatorname{spec}(A) \subset \Omega := \operatorname{int} K$, and let $f \in \mathcal{A}(K)$ be nonconstant with*

$$\|f\|_K = 1, \qquad |f| = 1 \text{ on } \partial K, \qquad \|f(A)\| = 2.$$

Then there are orthonormal vectors x, y such that, with $T = f(A)$,

$$Tx = 2y, \qquad T^*x = 0, \qquad T^*y = 2x, \tag{3.8}$$

and, for almost every $\sigma \in \partial K$,

$$\mathcal{P}(\sigma)^{1/2}\big(y - f(\sigma)x\big) = 0. \tag{3.9}$$

Proof. Apply the construction in Proposition 1. The spectral mapping theorem gives $\operatorname{spr}(T) < 1$, because the finitely many eigenvalues of A are inside Ω and a nonconstant inner function has modulus strictly less than 1 there. Choose x with $T^*Tx = 4x$, and put $y = Tx/2$. Lemma 2 gives $T^*x = 0$, $T^*y = 2x$, and $M_f\mathcal{V}x = \mathcal{V}y$. Here $\|y\| = 1$, and $2x^*y = x^*Tx = (T^*x)^*x = 0$, so x, y are orthonormal. The multiplication identity is exactly (3.9). □

Lemma 3 (Boundary eigenvalues reduce)**.** *If $Ax = \lambda x$, $\|x\| = 1$, and $\lambda \in \partial W(A)$, then $A^*x = \overline{\lambda}x$. Consequently, with $\simeq$ denoting unitary similarity,*

$$A \simeq A_\partial \oplus A_{\mathrm{in}},$$

where A_∂ is diagonal, with the boundary eigenvalues as its diagonal entries, while $\operatorname{spec}(A_{\mathrm{in}}) \subset \operatorname{int} W(A)$.

Proof. Choose a supporting direction ν, $|\nu| = 1$, at λ. Equality in the Rayleigh quotient gives

$$\operatorname{Re}(\overline{\nu}A)x = \operatorname{Re}(\overline{\nu}\lambda)x.$$

Substituting $Ax = \lambda x$ gives $A^*x = \overline{\lambda}x$. Thus $\mathbb{C}x$ reduces A. Split it off and repeat. Every eigenvalue belongs to $W(A)$, since an eigenvector realizes it, so the remaining spectrum is in the interior. □

Lemma 4 (Conformal and finite-interpolation facts)**.** *Let Ω be a bounded convex domain.*

1. *There is a conformal bijection $\phi : \Omega \to \mathbb{D}$, and it extends to a homeomorphism $\overline{\Omega} \to \overline{\mathbb{D}}$.*
2. *At each of finitely many distinct points of Ω, prescribe a contiguous jet, consisting of the value and all derivatives through a fixed finite order. If these data are realized by a function of $H^\infty(\Omega)$ of norm at most 1, they are also realized by $\mathcal{B} \circ \phi$, where $\mathcal{B}$ is a finite Blaschke product. If the infimum of the supremum norms over all interpolants of the prescribed data is 1, then the Schur-class interpolant is unique and is of this form.*

Proof. A bounded convex domain is a Jordan domain. The Riemann mapping theorem and the Carathéodory boundary theorem therefore give the first assertion.

Transfer the jet data to $\mathbb{D}$ by ϕ^{-1}. The confluent change-of-variables matrix between the original and transferred jets is triangular with nonzero diagonal, because $(\phi^{-1})' \neq 0$. Write the distinct disk nodes as ζ_i, prescribe derivatives through order $m_i - 1$, and let q be the Hermite polynomial with those jets. For a norm bound $\rho > 0$, the generalized Pick matrix used here is indexed by (i, r), $0 \le r < m_i$, and has entries

$$P(\rho)_{(i,r),(j,s)} = \frac{1}{r!s!} \partial_z^r \partial_{\overline{w}}^s \frac{\rho^2 - q(z)\overline{q(w)}}{1 - z\overline{w}} \bigg|_{z=\zeta_i,\, w=\zeta_j}. \tag{3.10}$$

This matrix depends only on the prescribed jets, not on the chosen Hermite representative. The finite confluent Nevanlinna–Pick theorem says that the data have an interpolant of norm at most ρ exactly when $P(\rho) \succeq 0$; see Sarason [13] and Bultheel–Lasarow [14]. At the norm bound $\rho = 1$, a positive definite matrix admits a Schur parametrization; choosing a unimodular terminal parameter gives a rational inner solution. In the scalar disk case, rational inner functions are finite Blaschke products. For general ρ, the corresponding conclusion is ρ times a rational inner function, obtained by rescaling the data. At the minimal norm, singularity yields the following uniqueness conclusion.

Contiguous jets at distinct nodes impose independent conditions: Hermite interpolation supplies polynomials $h_{i,r}$ whose normalized derivative functionals satisfy $h_{i,r}^{(s)}(\zeta_j)/s! = \delta_{ij}\delta_{rs}$. Thus there are no hidden linear dependencies in (3.10). Now let $\mathcal{J}$ send a holomorphic function on Ω to the prescribed jet vector $\mathbf{d}$, and put

$$c_0 = \inf\{\|g\|_\Omega : g \in H^\infty(\Omega),\ \mathcal{J}g = \mathbf{d}\}.$$

If $c_0 = 1$, then $P(1)$ is singular: if it were positive definite, continuity of (3.10) in ρ would give $P(\rho) \succeq 0$ for some $\rho < 1$, contradicting minimality. Let

$$\Theta(z) = \prod_i \left(\frac{z - \zeta_i}{1 - \overline{\zeta_i} z} \right)^{m_i}.$$

Two bounded holomorphic functions have the same prescribed jets exactly when their difference belongs to ΘH^∞. Thus the interpolation data form one coset of $H^\infty / \Theta H^\infty$, represented on the finite-dimensional model space $K_\Theta = H^2 \ominus \Theta H^2$ by an operator X commuting with the compressed shift. Sarason's lifting theorem identifies the least interpolant norm with $\|X\|$, so $\|X\| = 1$. The norm of X is attained because K_Θ is finite dimensional. Sarason's maximal-vector uniqueness theorem [13, Proposition 5.1] now gives a unique norm-one interpolant and says that it is inner. Its quotient formula uses two elements of the finite-dimensional rational space K_Θ, so the interpolant is rational and hence a finite Blaschke product. This also proves the uniqueness asserted in the lemma. The role of such finite-Blaschke extremizers in the matrix functional-calculus problem is developed by Crouzeix [1] and Li [9]. □

For a bounded domain Ω and a matrix A_0 with $\mathrm{spec}(A_0) \subset \Omega$, define

$$\psi_\Omega(A_0) = \sup\{\|g(A_0)\| : g \in H^\infty(\Omega),\ \|g\|_\Omega \le 1\}.$$

The value $g(A_0)$ depends only on finitely many derivatives, those prescribed by the Jordan blocks. Montel's theorem therefore makes the supremum a maximum. When Ω is convex, Lemma 4 replaces an extremizer by a finite Blaschke product composed with a Riemann map, without changing $g(A_0)$.

Lemma 5 (Strict domain monotonicity). *Let $\Omega_0 \subsetneq \Omega_1$ be bounded convex domains and suppose* $\operatorname{spec}(A_0) \subset \Omega_0$. *If $\psi_{\Omega_1}(A_0) > 1$, then*

$$\psi_{\Omega_0}(A_0) > \psi_{\Omega_1}(A_0).$$

Proof. Monotonicity in the displayed direction is immediate by restriction. Suppose equality held, with common value $\eta > 1$, and choose a norm-one extremizer g on Ω_1. Its norm on Ω_0 must be exactly 1, since otherwise rescaling its restriction gives a value larger than η. Thus it is also an extremizer on Ω_0.

For either domain, interpolate the finite jet of g that determines $g(A_0)$ with least possible norm. A norm smaller than 1 would contradict extremality after rescaling. The uniqueness part of Lemma 4 identifies g on each domain with a finite Blaschke product composed with its Riemann map. In particular, the restriction of g to Ω_0 has a continuous extension to $\overline{\Omega_0}$ with modulus one on $\partial\Omega_0$.

Choose $z_\star \in \Omega_0$. Along every ray from $z_\star$, convexity gives an exit radius for each domain. Since the domains differ, on some ray the first exit occurs strictly earlier; hence there is $\xi \in \partial\Omega_0 \cap \Omega_1$. At this interior point of Ω_1, continuity of the original holomorphic function and of the boundary extension gives $|g(\xi)| = 1$. The maximum-modulus principle makes g constant, which would give $\eta = 1$, a contradiction. □

4. Reduction to an interior-spectrum equality pair

Lemma 6 (Boundary-spectrum induction). *Assume Theorem 1 has been proved in every size smaller than N. Let $A \in M_N(\mathbb{C})$, let $K = W(A)$, and suppose $\psi(A) = 2$. Then either K is a nondegenerate closed disk, or*

$$\operatorname{spec}(A) \subset \operatorname{int} K. \tag{4.1}$$

Proof. By Lemma 1, K has nonempty interior. Choose polynomials p_j such that

$$\|p_j\|_K = 1, \qquad \|p_j(A)\| \longrightarrow 2. \tag{4.2}$$

Suppose A has a boundary eigenvalue. Lemma 3 gives $A \simeq A_\partial \oplus A_1$, where A_∂ is diagonal with spectrum on ∂K and $\operatorname{spec}(A_1) \subset \operatorname{int} K$. The A_∂-block of $p_j(A)$ has norm at most 1, so

$$\|p_j(A_1)\| \longrightarrow 2. \tag{4.3}$$

In particular A_1 is nonempty. Since $W(A_1) \subset K$, (4.3) and Proposition 1 give $\psi(A_1) = 2$. Its size is smaller than N, so induction says that

$$K_1 := W(A_1)$$

is a closed disk of positive radius.

Apply Lemma 3 once more, now to A_1 relative to K_1:

$$A_1 \simeq A_{\partial,1} \oplus A_2,$$

where $A_{\partial,1}$ is a finite diagonal block on ∂K_1 and $\operatorname{spec}(A_2) \subset \operatorname{int} K_1$. The same sequence (4.2) is bounded by 1 at the eigenvalues of $A_{\partial,1}$, all of which lie in K, so $\|p_j(A_2)\| \to 2$. Thus A_2 is nonempty.

Write $\Lambda_1 = \operatorname{spec}(A_{\partial,1})$. Since

$$K_1 = W(A_1) = \operatorname{conv}(\Lambda_1 \cup W(A_2)),$$

every extreme point of K_1 belongs to the compact set $\Lambda_1 \cup W(A_2)$. Indeed, a finite convex representation of an extreme point can contain no two distinct points with positive coefficients. The circle ∂K_1 has infinitely many extreme points and Λ_1 is finite. Hence $\partial K_1 \setminus \Lambda_1 \subset W(A_2)$; closedness and convexity of $W(A_2)$ give

$$W(A_2) = K_1. \tag{4.4}$$

Let $\Omega_0 = \operatorname{int} K_1$ and $\Omega_1 = \operatorname{int} K$. The polynomial sequence and Proposition 1 show

$$\psi_{\Omega_0}(A_2) = \psi_{\Omega_1}(A_2) = 2. \tag{4.5}$$

Indeed,

$$\|p_j\|_{\Omega_0} = \|p_j\|_{K_1} \le 1, \qquad \|p_j\|_{\Omega_1} = \|p_j\|_K = 1,$$

so the lower bounds follow from (4.2)–(4.4), while the upper bounds follow from the H^∞ estimate in Proposition 1. If $K_1 \subsetneq K$, then $\Omega_0 \subsetneq \Omega_1$, and Lemma 5 contradicts (4.5). Therefore $K_1 = K$, and K is the asserted disk. If this conclusion does not occur, A has no boundary eigenvalue, which is (4.1). □

Lemma 7 (The holomorphic extremizer)**.** *Suppose $K = W(A)$ has nonempty interior and $\operatorname{spec}(A) \subset \Omega = \operatorname{int} K$. If $\psi(A) = 2$, there is a nonconstant*

$$f = \mathcal{B} \circ \phi \in \mathcal{A}(K),$$

where $\phi : \Omega \to \mathbb{D}$ is conformal and $\mathcal{B}$ is a finite Blaschke product, such that

$$\|f\|_K = 1, \qquad |f| = 1 \text{ on } \partial K, \qquad \|f(A)\| = 2. \tag{4.6}$$

Proof. Normalized polynomials witnessing $\psi(A) = 2$ show $\psi_\Omega(A) \ge 2$, and Proposition 1 gives the reverse inequality. Montel compactness, followed by convergence of the finitely many spectral jets, gives an $H^\infty(\Omega)$ maximizer of norm at most one. Its norm is exactly one, since otherwise rescaling it would produce a matrix value of norm greater than two. Apply the finite Schur algorithm in Lemma 4 to the jet that determines its matrix value. It supplies $\mathcal{B} \circ \phi$ with the same matrix value. The boundary extension of ϕ puts this function in $\mathcal{A}(K)$ and gives boundary modulus 1. It cannot be constant because a constant matrix has norm at most 1. □

5. From equality to a rational inner function

Lemma 8 (A curved numerical-range arc)**.** *If $K = W(A)$ has interior, $\operatorname{spec}(A) \subset \operatorname{int} K$, and K is not a polygon, then ∂K contains an open real-analytic arc on which every supporting line exposes one point and the curvature is positive. Moreover, under $\operatorname{spec}(A) \subset \operatorname{int} K$, K cannot be a polygon.*

Proof. Put

$$\mathcal{H}(\theta) = \operatorname{Re}(e^{-i\theta} A), \qquad h_A(\theta) = \lambda_{\max}(\mathcal{H}(\theta)).$$

By Rellich's analytic perturbation theorem [15], for every θ_0 the eigenvalues of the real-analytic Hermitian matrix family $\mathcal{H}(\theta)$ can be numbered, with multiplicity, by real-analytic functions of θ on a neighborhood of θ_0. Take a finite cover of the parameter circle by such intervals. On each interval, place two branches in the same class when they agree on a nonempty open subinterval; the analytic identity theorem then makes them identical on the whole interval. Choose one representative of each class and shrink slightly to a compact subinterval. The difference of any two distinct representatives is a nonzero real-analytic function and therefore has only finitely many zeros there. Subdividing at the union of these finitely many crossing points fixes the ordering of all branches; on every resulting open interval their maximum h_A is one analytic branch. A finite cover and a common refinement give the asserted finite subdivision of the parameter circle. This is the finite-dimensional support-function form of the analytic Kippenhahn-curve description; compare [17].

At a differentiability point, the exposed boundary point is

$$\sigma(\theta) = e^{i\theta}\big(h_A(\theta) + ih'_A(\theta)\big), \qquad \sigma'(\theta) = ie^{i\theta}\big(h_A(\theta) + h''_A(\theta)\big). \tag{5.1}$$

The first identity follows by differentiating the top Rayleigh quotient; the second follows by differentiation. Convexity gives $h_A + h''_A \geq 0$. If this function vanished identically on every analytic interval, σ would be constant on each interval and K would be a polygon. Otherwise analyticity supplies a subinterval on which $h_A + h''_A > 0$. Formula (5.1) then gives the required regular, positively curved arc. Differentiability of the support function makes the exposed face a singleton there.

It remains to exclude a polygon when the spectrum is interior. Let λ be a vertex and let a unit vector v realize $v^*Av = \lambda$. Two linearly independent supporting normals ν_1, ν_2 at the vertex give

$$\mathrm{Re}(\overline{\nu_j}A)v = \mathrm{Re}(\overline{\nu_j}\lambda)v \quad (j = 1, 2).$$

The two independent real-linear equations imply $Av = \lambda v$ and $A^*v = \overline{\lambda}v$. Thus λ is a boundary eigenvalue, contradicting $\mathrm{spec}(A) \subset \mathrm{int}\, K$. $\square$

Lemma 9 (Boundary-arc uniqueness)**.** *Let $D \subset \mathbb{C}$ be a connected domain and let $\Gamma_0 \subset \partial D$ be a nonempty open regular real-analytic arc such that, near each point of Γ_0, D is exactly one of the two local components cut out by the arc. If g is holomorphic on D, continuous on $D \cup \Gamma_0$, and its boundary values vanish arclength-almost everywhere on Γ_0, then $g \equiv 0$ on D.*

Proof. Every open subarc has positive arclength, so continuity first makes g zero at every point of Γ_0. Near one such point, complexify a regular real-analytic parametrization of the arc. Its derivative is nonzero, so the holomorphic inverse-function theorem gives a biholomorphic coordinate that sends the arc to a real interval and the local part of D to one half-disk. In that coordinate, extend g by zero to the opposite half-disk. The extension is continuous; subdividing any triangle along the diameter and applying Cauchy's theorem on the pieces proves Morera's criterion. The extension is holomorphic and zero on an open half-disk, hence zero throughout the disk by the identity theorem. Thus g vanishes on a nonempty open subset of D, and connectedness gives the conclusion. $\square$

Proposition 3 (Rational collapse)**.** *Under the hypotheses and notation of Lemma 7, let x, y be the equality vectors from Proposition 2. Then the function f is rational on Ω. More precisely, for $z \notin \mathrm{spec}(A)$, set*

$$a(z) = x^*(zI - A)^{-1}x, \qquad c(z) = y^*(zI - A)^{-1}x,$$

and regard a, c thereafter as their meromorphic adjugate continuations. Then

$$c(z)f(z) = 2a(z) \tag{5.2}$$

as an identity of meromorphic functions on Ω. Equivalently, $f = 2a/c$ wherever the quotient is defined, and then everywhere after removable cancellation. After cancellation, $f = U/V$ for coprime polynomials with

$$\deg U > \deg V. \tag{5.3}$$

If adj *denotes the classical adjugate and*

$$D_A(z) = x^*\,\mathrm{adj}(zI - A)x, \qquad C_A(z) = y^*\,\mathrm{adj}(zI - A)x,$$

then the reduction can be chosen so that

$$U(z)C_A(z) = 2D_A(z)V(z).$$

Proof. Choose the arc from Lemma 8; it avoids the finite spectrum. For almost every point σ of that arc, let $\mathfrak{n}$ be its outward normal, $R = (\sigma I - A)^{-1}$, and

$$\Delta_\sigma = \mathrm{Re}\big(\overline{\mathfrak{n}}(\sigma I - A)\big) \succeq 0.$$

Equations (3.5) and (3.9) imply

$$\Delta_\sigma R\big(y - f(\sigma)x\big) = 0. \tag{5.4}$$

Indeed, squaring the norm in (3.9) and using $\mathcal{P}(\sigma) = \pi^{-1} R^* \Delta_\sigma R$ first gives $\Delta_\sigma^{1/2} R(y - f(\sigma)x) = 0$, and hence (5.4). The vector $r_\sigma = R(y - f(\sigma)x)$ is nonzero: R is invertible, while the orthonormality of x, y makes $y - f(\sigma)x \neq 0$. It is therefore an eigenvector of $\mathrm{Re}(\overline{\mathfrak{n}}A)$ for its maximal eigenvalue. The exposed point is unique, so $r_\sigma^* A r_\sigma = \sigma \|r_\sigma\|^2$. Since $(\sigma I - A) r_\sigma = y - f(\sigma)x$, taking the inner product with r_σ and conjugating gives

$$\big(y - f(\sigma)x\big)^* R\big(y - f(\sigma)x\big) = 0. \tag{5.5}$$

For $z \notin \mathrm{spec}(A)$, put $R_z = (zI - A)^{-1}$. This resolvent commutes with $T = f(A)$, and relations (3.8) give

$$x^* R_z y = \tfrac{1}{2} x^* R_z T x = \tfrac{1}{2} x^* T R_z x = 0, \qquad y^* R_z y = \tfrac{1}{2} y^* R_z T x = \tfrac{1}{2} y^* T R_z x = a(z). \tag{5.6}$$

Expanding (5.5), using $|f(\sigma)| = 1$ and (5.6), yields

$$2a(\sigma) - f(\sigma)c(\sigma) = 0 \tag{5.7}$$

almost everywhere on the arc. The punctured domain $D = \Omega \setminus \mathrm{spec}(A)$ is connected, and $cf - 2a$ is holomorphic on D and continuous up to the chosen arc. Convexity supplies the local one-sidedness required by Lemma 9; that lemma gives $cf - 2a = 0$ on D, hence as a meromorphic identity on Ω. Since $a(z) = z^{-1} + O(z^{-2})$ at infinity, a is not the zero rational function; the identity therefore also shows that c is not identically zero. This proves (5.2).

The adjugate formulas are

$$a = \frac{D_A}{\det(zI - A)}, \qquad c = \frac{C_A}{\det(zI - A)}.$$

The polynomial D_A has degree exactly $N - 1$, with leading coefficient 1. Since $x^* y = 0$, the degree of C_A is at most $N - 2$. Let G be a greatest common divisor of C_A and D_A, and choose

$$U = \frac{2D_A}{G}, \qquad V = \frac{C_A}{G}.$$

Then U, V are coprime, $f = U/V$, and $UC_A = 2D_A V$. Canceling G preserves the strict degree difference, proving (5.3). □

6. The full lemniscate is the boundary

Lemma 10 (Zeros of the transfer function)**.** *Let $f = U/V$ be the reduced rational function from Proposition 3. Then every finite zero of U lies in Ω, every zero of V lies outside K, and $f(\infty) = \infty$.*

Proof. For $z \notin K$, put $R = (zI - A)^{-1}$. Then $a(z) \neq 0$. Indeed, if $a(z) = x^* R x = 0$ and $v = Rx$, then $v \neq 0$ and $(zI - A)v = x$. Since $x^* v = a(z) = 0$, conjugation gives $v^* x = 0$, and hence

$$v^*(zI - A)v = v^* x = 0, \qquad \text{hence} \qquad \frac{v^* A v}{\|v\|^2} = z,$$

which would put z in $W(A) = K$, a contradiction.

By the construction in Proposition 3, $U = 2D_A/G$, so every zero of U is a zero of D_A. Since $\det(zI - A) \neq 0$ off K and $a = D_A/\det(zI - A)$ does not vanish there, U has no zero off K. Nor can V vanish at a boundary point: by coprimality, U/V would then be unbounded along approaches from Ω, where it equals the original function $f \in \mathcal{A}(K)$. Thus U/V extends continuously to ∂K, agrees there with f, and has modulus one. In particular, U has no boundary zero, so all its zeros lie in Ω. A reduced pole cannot lie in Ω, where f is holomorphic. Hence all poles lie outside K. Finally (5.3) says exactly that $f(\infty) = \infty$. $\square$

Proposition 4 (Full-level identity)**.** *For the reduced rational extremizer $f = U/V$,*

$$\{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| < 1\} = \Omega, \qquad \{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| = 1\} = \partial K. \quad (6.1)$$

Moreover f' does not vanish on ∂K, and ∂K is a smooth real-analytic strictly convex Jordan curve.

Proof. Put

$$\mathcal{O} := \{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| < 1\}.$$

No pole belongs to $\overline{\mathcal{O}}$, because $|f| > 1$ in a punctured neighborhood of each pole. If $z \in \partial\mathcal{O}$, continuity and approximation from $\mathcal{O}$ give $|f(z)| \leq 1$; strict inequality would put a neighborhood of z in $\mathcal{O}$. Therefore

$$\partial\mathcal{O} \subset \{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| = 1\}. \quad (6.2)$$

Now $\Omega \subset \mathcal{O}$, while ∂K is disjoint from $\mathcal{O}$. Hence the connected set Ω is both open and closed relative to $\mathcal{O}$, and is one of its components.

Every component $\mathcal{O}_0$ is bounded because $f(\infty) = \infty$, and it contains a zero of f. Otherwise its compact closure would avoid all zeros of f: any zero in the closure lies in $\mathcal{O}$ and hence in $\mathcal{O}_0$. Thus $1/f$ would be holomorphic near $\overline{\mathcal{O}_0}$, have modulus one on the boundary by (6.2), and have modulus greater than one inside, contrary to the maximum-modulus principle. Lemma 10 puts every zero in Ω, so $\mathcal{O}$ has no other component. This proves the first identity in (6.1).

Conversely, if $V(z_0) \neq 0$ and $|f(z_0)| = 1$, the open mapping theorem says that every neighborhood of z_0 contains points where $|f| < 1$. Thus $z_0 \in \partial\mathcal{O}$, and (6.2) proves the second identity in (6.1).

Suppose that $f'(z_0) = 0$ at a level point. Then, in a neighborhood of z_0,

$$f(z) - f(z_0) = (z - z_0)^k q(z), \qquad k \geq 2, \quad q(z_0) \neq 0.$$

Choose $\alpha \in \mathbb{C}$ so that $\operatorname{Re}\big(\overline{f(z_0)}\alpha^k q(z_0)\big) < 0$, and let $\zeta = e^{2\pi i/k}$. For all sufficiently small $t > 0$, the points $z_j = z_0 + t\alpha\zeta^j$, $0 \leq j < k$, are not poles and satisfy

$$|f(z_j)|^2 = 1 + 2t^k \operatorname{Re}\big(\overline{f(z_0)}\alpha^k q(z_0)\big) + O(t^{k+1}) < 1.$$

The first identity in (6.1) puts every z_j in Ω. But

$$\frac{1}{k}\sum_{j=0}^{k-1} z_j = z_0$$

because $\sum_{j=0}^{k-1} \zeta^j = 0$. Convexity of Ω would then put the boundary point z_0 in Ω, a contradiction. Hence $f' \neq 0$ on the level, and the implicit-function theorem makes the boundary real analytic and smooth.

Finally suppose that the boundary contained a nontrivial segment of a real line $z = z_0 + t\xi$, $t \in \mathbb{R}$, $\xi \neq 0$. The expression

$$|U(z_0 + t\xi)|^2 - |V(z_0 + t\xi)|^2$$

is a polynomial in the real variable t. Its vanishing on an interval would make it vanish for every real t. If $V(z_0 + t\xi) = 0$ for some real t, the same identity would force $U(z_0 + t\xi) = 0$, contrary to coprimality. The full-level identity would therefore put the entire unbounded line in ∂K, contradicting compactness. Thus the boundary has no nontrivial segment, which for a compact convex body is exactly strict convexity. □

7. The algebraic tangent argument

The Hermitian determinant below is the line-coordinate polynomial introduced by Kippenhahn [16]; modern projective-dual accounts appear in [17]. Throughout this section, primal projective coordinates are ordered as $[Z : X : Y]$, dual coordinates as $[s : u : v]$, and incidence is

$$sZ + uX + vY = 0.$$

Thus the affine point $z = X + iY$ is $[1 : X : Y]$.

Proposition 5 (Generic hyperbolic contact count)**.** *Let $\mathscr{C} \subset \mathbb{P}^2_{\mathbb{C}}$ be an irreducible projective curve defined over $\mathbb{R}$, and suppose its entire real locus is a smooth strictly convex oval in the affine chart $Z = 1$. Let $\mathscr{Q}(s, u, v)$ be a real irreducible homogeneous equation of the dual curve $\mathscr{C}^*$. If*

$$\mathscr{Q}(1, 0, 0) \neq 0 \quad \textit{and} \quad s \longmapsto \mathscr{Q}(s, u, v) \textit{ has only real zeros for all } (u, v) \in \mathbb{R}^2,$$

then $\deg \mathscr{Q} = 2$.

Proof. Write $m = \deg \mathscr{Q}$. The normalization, conormal, and biduality theory of projective plane curves in characteristic zero gives the following generic contact description; see Tevelev [20].

1. Outside a finite subset $B \subset \mathscr{C}^*$, a smooth dual point has one smooth contact point on $\mathscr{C}$, and the tangent line to $\mathscr{C}^*$ at that dual point recovers the contact point. This is the generic one-to-one Gauss correspondence supplied by biduality; B absorbs the singular, branch, and ramification loci.
2. Since $[1 : 0 : 0] \notin \mathscr{C}^*$, projection from that point,
$$\pi : \mathscr{C}^* \longrightarrow \mathbb{P}^1, \qquad [s : u : v] \longmapsto [u : v],$$
is everywhere defined and has generic degree m: a generic fiber is the intersection with a line through $[1 : 0 : 0]$, and Bézout gives m points.
3. In characteristic zero this projection is separable. Equivalently, the discriminant of $\mathscr{Q}(s, u, v)$, viewed as a binary homogeneous polynomial in (u, v), is not identically zero. The branch values of π and the projection of B therefore form a finite exceptional subset of $\mathbb{P}^1$.
4. Choose a real $[u : v]$ outside that set. Because the coefficient of s^m is $\mathscr{Q}(1, 0, 0) \neq 0$, specialization does not lower the degree. Hence there are m distinct roots s_j, all real by hypothesis, and $\ell_j = [s_j : u : v]$ are smooth real points of $\mathscr{C}^* \setminus B$.
5. Biduality and the chosen coordinate order identify the unique contact point with
$$[Z : X : Y] = [\mathscr{Q}_s(\ell_j) : \mathscr{Q}_u(\ell_j) : \mathscr{Q}_v(\ell_j)].$$
The polynomial and ℓ_j are real, so this point is real and therefore lies on the given oval.
6. A fixed unoriented normal $[u : v]$ has exactly two tangent lines to a smooth strictly convex oval: the unique support lines at the maximum and minimum of $uX + vY$. Thus the m lines just found give $m \leq 2$. Conversely, both support lines are points of $\mathscr{C}^*$, so they are two distinct roots and $m \geq 2$. Hence $m = 2$.

□

Proposition 6 (Circle criterion)**.** *Let K be a compact convex body whose boundary is the full level set*

$$\partial K = \{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| = 1\}, \tag{7.1}$$

where U, V are coprime polynomials, $\deg U > \deg V$, K is strictly convex, and $(U/V)'$ does not vanish on ∂K. Writing $z = X + iY$, suppose there are Hermitian matrices H, J such that, in the real dual coordinates $[s : u : v]$ associated with the line $s + uX + vY = 0$, every tangent line to an open boundary arc belongs to the projective curve

$$\det(sI + u\mathsf{H} + v\mathsf{J}) = 0. \tag{7.2}$$

Then ∂K is a circle.

Proof. Put $f = U/V$. Write $z = X + iY$, $w = X - iY$, and $S^{\#}(w) = \overline{S(\overline{w})}$ for a polynomial S. For a homogenization $S_h(w, Z)$, the notation $S_h^{\#}$ means coefficientwise conjugation, with w, Z left unchanged. Put $\mathbf{z} = (Z, X, Y)^{\mathsf{T}}$ and $\boldsymbol{\ell} = (s, u, v)^{\mathsf{T}}$, in the coordinate order fixed above. Let $d = \deg U$, $e = \deg V < d$, and homogenize the real lemniscate polynomial to degree $2d$:

$$\mathscr{F}(Z, X, Y) = U_h(X + iY, Z)U_h^{\#}(X - iY, Z) - Z^{2(d-e)}V_h(X + iY, Z)V_h^{\#}(X - iY, Z). \tag{7.3}$$

Here U_h, V_h are the usual homogenizations to their own degrees. At $Z = 1$, use the Wirtinger derivative $\partial_z = \frac{1}{2}(\partial_X - i\partial_Y)$. Then

$$\mathscr{F} = |V|^2(|f|^2 - 1), \qquad \partial_z \mathscr{F} = |V|^2 f'\overline{f} \quad \text{on } |f| = 1.$$

Here $V \neq 0$ on the level and $f' \neq 0$ there by hypothesis, so the boundary lies in the regular locus of $\mathscr{F} = 0$.

Selection of the real component. Factor $\mathscr{F}$ over $\mathbb{C}$. At a regular zero exactly one irreducible factor occurrence vanishes: two vanishing factors, or a repeated vanishing factor, would make every first derivative of $\mathscr{F}$ zero. The choice of that factor is locally constant along the boundary, because all other factors remain nonzero nearby. The boundary is connected, so one factor $\mathscr{P}$ vanishes on all of it; let $\mathscr{C} = V(\mathscr{P})$. Since $\mathscr{F}$ has real coefficients, the coefficientwise conjugate $\overline{\mathscr{P}}$ is also a factor and vanishes at each real boundary point. Local uniqueness makes $\overline{\mathscr{P}}$ an associate of $\mathscr{P}$; rescaling $\mathscr{P}$ therefore gives real coefficients. Thus $\mathscr{C}$ is an irreducible real projective curve, and it is smooth along the boundary.

The entire real locus. Every real affine point of $\mathscr{C}$ is a real zero of (7.3). If $V(z) = 0$ there, (7.3) also forces $U(z) = 0$, contradicting coprimality. Hence $V(z) \neq 0$, and (7.1) puts the point on ∂K. The reverse inclusion holds by construction of $\mathscr{C}$, so

$$\mathscr{C}(\mathbb{R}) \cap \{Z \neq 0\} = \{[1 : X : Y] : X + iY \in \partial K\}.$$

There are no additional real points at infinity. Indeed, if $\mathrm{lc}(U)$ denotes the nonzero leading coefficient of U,

$$\mathscr{F}(0, X, Y) = |\,\mathrm{lc}(U)|^2(X^2 + Y^2)^d, \tag{7.4}$$

and $X^2 + Y^2 > 0$ for every nonzero real projective pair $[X : Y]$. Therefore

$$\mathscr{C}(\mathbb{R}) = \{[1 : X : Y] : X + iY \in \partial K\},$$

exactly one smooth strictly convex affine oval.

Let $\mathscr{C}^*$ be the projective dual curve, the Zariski closure of the tangent lines at smooth points of $\mathscr{C}$, and let $\mathscr{Q}(s, u, v)$ be its irreducible homogeneous equation, chosen with real coefficients. Tangency on an open real arc and (7.2) imply

$$\mathscr{Q}(s, u, v) \text{ divides } \det(sI + u\mathsf{H} + v\mathsf{J}). \tag{7.5}$$

Indeed, the Gauss image of that arc is nonconstant: a constant tangent line would contain a nontrivial boundary arc, contrary to strict convexity. Its image is therefore infinite and hence Zariski dense in the irreducible curve $\mathscr{C}^*$. The determinant vanishes on that dense set and thus on all of $\mathscr{C}^*$. The homogeneous prime ideal of a plane curve is the principal ideal $(\mathscr{Q})$, proving (7.5). Since $\mathscr{C}$ is real, its dual is invariant under conjugation; as in the primal factor argument, $\mathscr{Q}$ can be rescaled to have real coefficients.

Set $\mathbf{e} = [1 : 0 : 0]$. The determinant in (7.5) equals 1 at $\mathbf{e}$, so $\mathscr{Q}(1,0,0) \neq 0$. If $m = \deg \mathscr{Q}$, homogeneity gives

$$[s^m]\mathscr{Q}(s,u,v) = \mathscr{Q}(1,0,0) \neq 0.$$

Thus specialization never lowers the degree in s. Specializing the polynomial divisibility (7.5) shows that every zero of $\mathscr{Q}(s,u,v)$ is a zero of the Hermitian determinant. For real u, v, the latter zeros are the negatives of the eigenvalues of $u\mathsf{H} + v\mathsf{J}$, and are therefore real. Hence $\mathscr{Q}$ is hyperbolic with respect to $\mathbf{e}$ in the elementary real-rootedness sense [18,19].

The exact real-locus identity above and Proposition 5 now give

$$\deg \mathscr{Q} = 2. \tag{7.6}$$

Thus $\mathscr{C}^*$ is an irreducible conic and is nonsingular. Writing its equation as $\boldsymbol{\ell}^\mathsf{T}\Sigma\boldsymbol{\ell} = 0$, with an invertible symmetric 3×3 matrix Σ, its tangent at $\boldsymbol{\ell}$ corresponds to the primal point $\Sigma\boldsymbol{\ell}$. The envelope of those tangents is $\mathbf{z}^\mathsf{T}\Sigma^{-1}\mathbf{z} = 0$, also a conic. Since it contains the generic bidual contact points, a Zariski-dense subset of $\mathscr{C}$, it is $\mathscr{C}$.

Finally, because the defining polynomial of $\mathscr{C}$ divides $\mathscr{F}$, every point of this conic at infinity must be one of

$$[0 : 1 : i], \qquad [0 : 1 : -i].$$

The line $Z = 0$ is not a component of $\mathscr{C}$: the curve is irreducible and contains its affine oval. Bézout's theorem therefore gives total intersection multiplicity two with the line at infinity. Complex conjugation exchanges the two displayed, distinct circular points and preserves multiplicity, so each occurs once. The real homogeneous equation of the conic consequently has the form

$$q_0(X^2 + Y^2) + q_1 ZX + q_2 ZY + q_3 Z^2 = 0, \qquad q_0 \neq 0.$$

Completing squares shows that its nonempty compact real oval is a circle. □

8. Proof of the main theorem

Proof of Theorem 1. We use induction on N. For $N = 1$, and more generally whenever the numerical range has empty interior, Lemma 1 gives $\psi(A) = 1$, so the hypothesis never occurs.

Assume the theorem holds for all sizes smaller than N, and let $A \in M_N(\mathbb{C})$ satisfy $\psi(A) = 2$. Put $K = W(A)$. It has nonempty interior. By Lemma 6, either K is already a nondegenerate disk or $\operatorname{spec}(A) \subset \Omega = \operatorname{int} K$. Consider the second case.

Choose $f \in \mathcal{A}(K)$ as in Lemma 7, and let x, y be the equality vectors from Proposition 2. Since an interior-spectrum numerical range cannot be a polygon, Lemma 8 provides a curved analytic boundary arc. On that arc, Proposition 3 yields

$$f(z) = \frac{U(z)}{V(z)}, \qquad \deg U > \deg V,$$

after cancellation of the meromorphic transfer identity (5.2).

Lemma 10 puts every zero of U in Ω, every pole outside K, and gives $f(\infty) = \infty$. Proposition 4 then gives

$$\{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| < 1\} = \Omega, \qquad \{z \in \mathbb{C} : V(z) \neq 0,\ |U(z)/V(z)| = 1\} = \partial K,$$

and shows that this boundary is a smooth strictly convex rational lemniscate.

Put $H_A = \operatorname{Re} A$ and $J_A = \operatorname{Im} A$. Writing $z = X + iY$, every supporting line $s + uX + vY = 0$, with $s, u, v \in \mathbb{R}$, can be oriented so that $sI + uH_A + vJ_A \succeq 0$. A unit vector attaining the point of support has zero quadratic form for this positive semidefinite matrix and therefore lies in its kernel. Hence

$$\det(sI + uH_A + vJ_A) = 0.$$

All hypotheses of Proposition 6 now hold. It follows that ∂K is a circle. Compactness and convexity make K the filled closed disk bounded by that circle.

The disk has positive radius because K has nonempty planar interior. Writing its center and radius as γ and ρ gives

$$W(A) = \{z : |z - \gamma| \le \rho\}, \qquad \rho > 0,$$

which completes the induction and the proof. $\square$

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